Amit Kumar
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Objective

Postdoctoral researcher developing adaptive biomaterials and environmentally responsive soft robotic systems through bioinspired material design, 4D printing, and stimuli-responsive polymers for biomedical and agricultural applications.

Research Area

My research area includes:

- **Functional Responsive Materials:** Hydrogel-, elastomer-, and nanoparticle-based systems with solvent and light programmable actuation.
- **Interfacial Polymer Chemistry:** Engineering dynamic biomaterials with biocompatible crosslinking strategies, bilayer architectures, and particle self-assembly to develop a novel multimaterial system for medical applications.
- **3D/4D Printing of Soft Materials:** Direct ink writing, bath-assisted printing, and molding of architectures that transform over time under environmental or external stimuli.
- **Soft Robotics:** Designing biodegradable, adaptive soft robotic systems for autonomous actuation and controlled molecular delivery in biomedical and agricultural applications.

Education

- **Doctor of Philosophy (Ph.D.) + Master of Science (M.S.)- 9.75 (CGPA)** July'17 - July'24
  Dual Degree Interdisciplinary Research Program,
  Department of Biotechnology, Indian Institute of Technology Madras, Chennai, India

  Thesis: *Harnessing Solvent-Responsive Actuation Behavior of Polymers for Biomedical Applications*
  Supervisors: *Prof. Nitish R. Mahapatra, Department of Biotechnology,*
  Prof. Pijush Ghosh, Department of Applied Mechanics & Biomedical Engineering

- **Bachelor of Technology (B.Tech.)- 70.42 (CPI)** Aug'13 - May'17
  Department of Biotechnology, Delhi Technological University, Delhi, India

  Major Project: *Review report on biosensors for detection of plant toxins*
  Supervisor: *Prof. Bansi D. Malhotra, Department of Biotechnology*

Professional Experience

- **Postdoctoral Researcher**, Istituto Italiano di Tecnologia (IIT), Italy 1st June'25 - Present
  (Mentor: Prof. Barbara Mazzolai, Director, Bioinspired Soft Robotics (BSR) Laboratory, IIT, Genoa)
  - **Hydrogel Powered Root Inspired Growing Robot (HyGROWBOT)**- Designing biodegradable hydrogel–elastomer composite actuators that can sense (pH, temperature) and move through the soil while enabling controlled release of bioactive molecules (nutrients, antimicrobial agents) in response to environmental cues—supporting sustainable soil health management.
  - **Sunlight-triggered Marangoni-powered locomotion on Water**- Designing a multiresponsive actuator for on-demand environment-powered locomotion on the water.
  - **Micropatterned surface for Superhydrophobic/Hydrophilic transition for Wound Dressing Application**- Developing multilayered chitosan-based wound dressing with waterproofing ability by changing from hydrophilic to superhydrophobic surface.

- **Research Associate**, Indian Institute of Technology Madras (IITM), India Sept'24 - Feb'25
  (Mentor: Prof. Pijush Ghosh, Nanomechanics Lab, Department of Applied Mechanics & Biomedical Engineering, IITM, Chennai)
  - Engineered dual solvent- and light-responsive bilayer polymer films with programmable actuation and validated their efficacy in cyclic biofilm removal, advancing smart antifouling biomaterials.

- **Industrial Trainee**, National Physical Laboratory (NPL), CSIR, Delhi, India — Jun'16 - Aug'16

(Mentor: Dr. G. Sumana Gajjala, Biomolecular Electronics and Conducting Polymer Research Group, NPL, CSIR)

  — Worked on FTIR, Zeta-Sizer, Electrophoretic Deposition Technique, and Electrochemical Analyzer.

  — Used Langmuir-Blodgett Film Deposition Technology for making a Monolayer of Molybdenum Di-Sulfide and Gold Nanoparticles (AuNPs) composite.

- **Industrial Trainee**, INMAS, DRDO, Delhi, India — Dec'15 - Jan'16

(Mentor: Dr. Aseem Bhatnagar, Department of Nuclear Medicine, INMAS, DRDO)

  — Assisted in making Melatonin-Caffeine Tablets for clinical trials that act as Radio-Protector.

  — Learned about the working of tablet punching machine, disintegration tester, radiolabelling with technisium, thin layer chromatography, gamma scintigraphy scan, CAPRAC-t, UV-IR spectrometer.

Research Publications

1. **Amit Kumar**[1], Shaik Sameer Basha[1], Mukilarasi B, Vimalraj Selvaraj, Pijush Ghosh, Swathi Sudhakar. Leveraging cancer microenvironment pH shift for actuation-induced multidrug release from chitosan/carboxymethyl cellulose bilayer film. *International Journal of Biological Macromolecules*, **(2025)**. DOI: 10.1016/j.ijbiomac.2025.144701 **(IF- 8.7)**

   — Engineered pH-responsive chitosan/carboxymethyl cellulose bilayer films with tunable actuation by tailoring solvent conditions to exploit the acidic tumor microenvironment.

   — Established curvature-mediated release of multiple anticancer drugs and validated therapeutic efficacy through in vitro studies, demonstrating the potential of actuation-assisted localized cancer therapy.

2. **Amit Kumar**, Smruti Parimita, Kumari Kiran, Nitish R. Mahapatra, Pijush Ghosh. Superhydrophobic Asymmetric pH-responsive Soft Actuators: Implications for the Development of Anti-fouling Medical Devices. *Chemical Engineering Journal*, **(2024)**. DOI: 10.1016/j.cej.2024.154772 **(IF- 12.5)**

   — Developed asymmetric superhydrophobic chitosan-based soft actuators using self-assembled silane-modified silica nanoparticles to achieve programmable solvent and pH-responsive deformation.

   — Demonstrated simultaneous antifouling, antibacterial, and 4D shape-morphing functionalities, enabling multifunctional soft materials for next-generation biomedical devices.

3. Nidhi Gupta[1], **Amit Kumar**[1], Pravin K Vaddavalli, Nitish R. Mahapatra, Akhil Varshney, Pijush Ghosh. Efficient Reduction of the Scrolling of Descemet Membrane Endothelial Keratoplasty Grafts by Engineering the Medium. *Acta Biomaterialia*, **(2023)**. DOI: 10.1016/j.actbio.2023.09.024 **(IF- 10.4)**

   — Elucidated the diffusion-driven deformation mechanism underlying DMEK graft scrolling and developed biomimetic chitosan bilayer models to investigate graft mechanics.

   — Engineered solvent-mediated strategies to regulate graft curvature, providing a simple and clinically translatable approach to improve corneal transplantation procedures.

4. **Amit Kumar**, Raja Rajamanickam, Joyita Hazra, Nitish R. Mahapatra, Pijush Ghosh. Engineering the Nonmorphing Point of Actuation for Controlled Drug Release by Hydrogel Bilayer across the pH Spectrum. *ACS Applied Materials & Interfaces*, **(2022)**. DOI: 10.1021/acsami.2c16658 **(IF- 7.8)**

   — Developed biocompatible chitosan/carboxymethyl cellulose hydrogel bilayers with programmable non-morphing interfaces for precise control of pH-responsive actuation.

   — Demonstrated bidirectional actuation-driven drug release and bioinspired deformation patterns, establishing a versatile platform for controlled therapeutic delivery.

5. Smruti Parimita, **Amit Kumar**, Hariharan Krishnaswamy, Pijush Ghosh. 4D printing of pH-responsive bilayer with programmable shape-shifting behaviour. *European Polymer Journal*. **(2024)**. DOI: 10.1016/j.eurpolymj.2024.113581 **(IF- 6.8)**

   — Fabricated 4D-printed pH-responsive (chitosan/carboxymethyl cellulose hydrogel) bilayer structures exhibiting programmable and reversible shape transformations under physiological conditions.

6. Smruti Parimita, **Amit Kumar**, Hariharan Krishnaswamy Pijush Ghosh. Solvent triggered shape morphism of 4D printed hydrogels. *Journal of Manufacturing Processes*, **(2023)**. DOI: 10.1016/j.jmapro.2022.11.065 **(IF- 7.8)**

   — Contributed to the fabrication of a printable hydrogel exhibiting programmable and reversible solvent-induced shape transformations.

7. Anas Saifi, Smruti Parimita, **Amit Kumar**, Pijush Ghosh. Programmable 4D-Printed Soft Actuators: Harnessing Bending Strain Distribution for Embedded Topological Functionality. *ACS Applied Engineering Materials*. **(2025)**. DOI: 10.1021/acsaenm.5c00225 **(IF- 3.5)**

   — Contributed to the development of programmable 4D-printed light responsive TPU elastomer via bath printing.

Research Publications (To Be Communicated/Under Review)

1. **Amit Kumar**, Anas Saifi, Kumari Kiran, Sanjib Senapati, Nitish R. Mahapatra, Pijush Ghosh. Dual-Responsive Bidirectional Actuator for Biofilm Disruption on Medical Surfaces. *Advanced Healthcare Materials*- **(Under Review)**

   — Developed a dual-responsive chitosan–PDMS bilayer soft actuator that integrates solvent- and green light–triggered orthogonal stimuli to achieve programmable, reversible, and bidirectional mechanical deformation for active biofilm disruption.

   — Demonstrated a non-chemical, mechanically driven strategy for biofilm removal via cyclic peeling and compression, offering a biocompatible and reusable platform for antifouling biomedical devices and self-cleaning implant interfaces.

2. Ruowen Tu[1], **Amit Kumar**[1], Barbara Mazzolai. An Application-Driven Framework for Designing and Assessing Biodegradable Soft Robots. *Perspective*- **(To be Submitted)**

   — The perspective provides a roadmap for designing biodegradable soft robots that integrates functionality with sustainability, rather than treating biodegradability as a standalone material property.

3. Yesheneh Jejaw Mamo, Ruowen Tu, **Amit Kumar**, Yasmin Tauqeer Ansari, Barbara Mazzolai, Nicola M. Pugno. Hyperelastic Hierarchical Origami Actuators Powered by Hydrogel Swelling. *Article*- **(To be Submitted)**

   — Contributed to the development of hydrogel sacrificial mold-based fabrication of 3D elastomer origami and demonstrated reversible hydrogel-powered actuation, providing a scalable strategy for programmable and reusable soft robotic systems.

Technical Skills

- **Laboratory Proficiencies:**
  - **Responsive biomaterials fabrication:** Hydrogels, Elastomers, Nanoparticles, Bilayers, 3D printing (DIW)
  - **Advanced microscopy and spectroscopy:** SEM, FT-IR, UV-Vis
  - **Cellular assays:** MTT, live/dead bacterial viability, antibacterial studies, biodegradability studies
- **Computational and Analytical Tools:**
  - **Visualization & Design tools:** Origin, Graphpad Prism, Chemdraw, Inkscape, Blender, Filmora 9.0, Ultimaker Cura
  - **Documentation:** Microsoft Office, LaTeX

Posters and Presentations

1. **Amit Kumar**, Smruti Parimita, Kumari Kiran, Nitish R. Mahapatra and Pijush Ghosh,"Superhydrophobic Asymmetric Solvent Responsive Actuator for Antifouling Medical Devices", *Applied Mechanics and Biomedical Engineering (AMBE) Research Palooza*, 5th July 2024, IIT Madras, Chennai, India.

2. **Amit Kumar**, Nidhi Gupta, Nitish R. Mahapatra and Pijush Ghosh, "Strategies to attain underwater actuation of solvent-responsive chitosan and its biomedical applications", *33rd Annual Conference of the European Society for Biomaterials* (ESB-2023), 4th – 8th September 2023, Davos, Switzerland.

3. **Amit Kumar**, V.P. Anju, Nitish R. Mahapatra and Pijush Ghosh, "Impact of Tuning Solvent Characteristics on Self-folding of Biopolymer Thin Film", *Active Polymeric Materials and Microsystems* (APMM-2019), 16th – 19th September 2019, Dresden, Germany.

4. **Amit Kumar**, Smruti Parimita, Nitish R. Mahapatra and Pijush Ghosh, "pH-Responsive Soft Actuator for Biomedical Applications", *3D Graphy Engineering and Medical 2023*, (3DGRAPHY- 2023), 9th - 10th December 2023, IIT Bombay, Maharashtra, India.

5. **Amit Kumar**, Nitish R. Mahapatra and Pijush Ghosh, "pH-responsive bilayer films with engineered non-morphing point of actuation for controlled model drug release", *14th International Conference on Advancements in Polymeric Materials* (APM-2023), 17th – 19th March 2023, CIPET (SARP)-APDDRL, India.

Workshop Attended

1. Tumor Microenvironment and Resistance to Drugs: The Role of Biomaterials, In Silico and In Vitro Modeling, 19^{th} September 2025, University of Pavia, Italy.

2. Additive Manufacturing of Bio-Implants - Academic & Industry Perspectives, 10^{th} - 11^{th} March 2023, Indian Institute of Technology Madras, India.

Research Awards and Grants

- **Marie Curie Postdoctoral Fellowship- Seal of Excellence** 2026
  - The proposal "4DEX-mplant" secured 90.80% and was recognized as a high-quality project proposal by the European Commission.
- **Research Excellence Award-** IIT Madras 2024
  - Best Research Publication in the Department of Biotechnology, awarded by IIT Madras during Dec 2023- Dec 2024.
- **Institute Research Award-** IIT Madras 2024
  - Best Doctoral Research in the Department of Biotechnology, awarded by IIT Madras.
- **First Prize-** Sci-Tech Exposium, IIT Madras 2024
  - First prize in oral presentation at Applied Mechanics and Biomedical Engineering (AMBE) Research Palooza.
- **Graduate Aptitude Test in Engineering (GATE)-** India 2017
  - National level scholarship exam to pursue M.S. (by research) from IIT Madras through HTRA scheme.

Position of Responsibility

- **Mentoring- PhD Scholar**, IIT Italy Dec'25 - Present
  - Provided PhD mentorship in designing hydrogel sacrificial mold-based 3D elastomer origami actuators and demonstrating reversible hydrogel swelling-driven actuation for programmable soft robotic systems.
- **Workshop Organiser- IEEE RAS ROBOSOFT 2026**, Kanazawa, Japan April'26
  - Title: *Functional and Untethered Soft Machines Powered by the Environment*
  - Theme of the workshop: Can soft machines achieve true autonomy by dynamically adapting to environmental changes or by harvesting and utilizing ambient energy sources?
  - The workshop included: Invited talks, a concept competition, a demonstration competition, and a poster competition.
- **Guest Editor**, Special Issue, Bioinspiration & Biomimetics, IOP Publishing Present
  - The Special Issue on the topic "Soft Robots and Actuators Powered by the Environment", seeks to consolidate and advance research at the intersection of soft robotics, bioinspiration, environmental energy harvesting, and sustainable design.
- **Teaching Assistant- Genetic Engineering Lab**, IIT Madras Aug'19 - Dec'19
  - Conducted competent cells, transformation, plasmid isolation, gel electrophoresis, restriction digestion, and PCR amplification experiments.
- **Teaching assistant- Microbiology Lab**, IIT Madras Aug'18 - Dec'18
  - Successfully conducted experiments for media preparation, pure culture isolation, Gram staining, and selective media preparation.
- **Peer Reviewer** Present
  - Reviewed 9 articles for RSC Advances(3), RSC Sustainability(2), Sustainable Food Technology(1), and ACS Omega(3).

Extracurricular Activities

- Secured 3rd place in both football and Pole vault (athletics) at Inter IIT Sports Meet, IIT Gandhinagar in December, 2023.
- Captained hostel football team to become Champions in 11v11 football tournament during Schroeter, an Annual Inter Hostel Sports Meet 2022-23 among 16 hostels.
- Held Inter IIT Meet Record holder for 3 years from 2018-2022 in Pole Vault with a height of 3.82m and won the Gold Medals for three consecutive year in Inter IIT Sports Meet among 21 IITs across India.
- Bagged 1st place in 8km Road Race, Dean's trophy 2017-18 held at IIT Madras.

Academic Referees

1. Prof. Barbara Mazzolai, Director, Bioinspired Soft Robotics Group, Italiano di Tecnologia (IIT), Genoa, Italy (email: barbara.mazzolai@iit.it)
2. Prof. Nitish R. Mahapatra, Professor, Department of Biotechnology, Bhupat and Jyoti Mehta School of Biosciences, Cardiovascular Genetics Lab, Indian Institute of Technology Madras, India (email: nmahapatra@iitm.ac.in)
3. Prof. Pijush Ghosh, Professor, Department of Applied Mechanics & Biomedical Engineering, Nanomechanics Lab, Indian Institute of Technology Madras, India (email: pijush@iitm.ac.in))
4. Dr. Nidhi Gupta, Senior Consultant, Dr. Shroff's Charity Eye Hospital, India (email: nidhiophthal@gmail.com)